

Jose

①

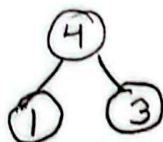
①

3 is greater or equal than parent?



3 swap

1 4 is $\geq$ parent?



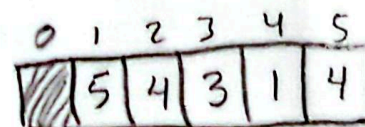
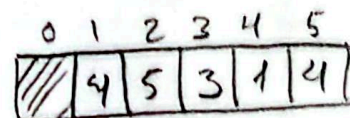
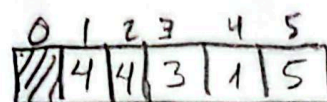
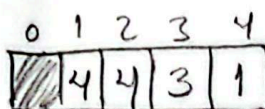
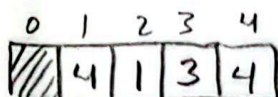
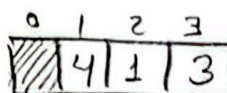
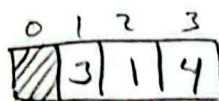
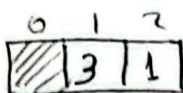
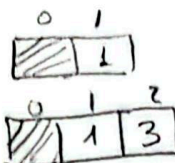
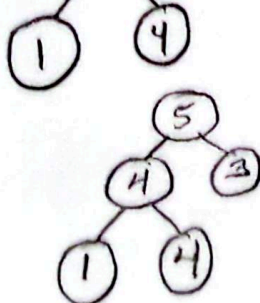
④

Diagram illustrating a swap operation in a binary tree. The root node is 1, and its left child is 4. An arrow labeled "swap" points from node 1 to node 4. Below node 4, the text "is > parent?" is written.



is $\geq$ parent?

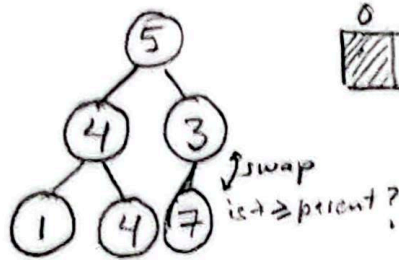
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graph TD; 4((4)) --- 5((5)); 4 --- 3((3));
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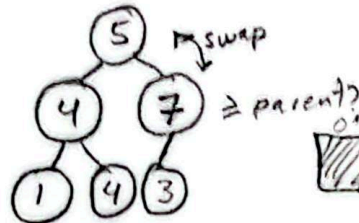
Q1. a) 1, 3, 4, 4, 5, 7, 9, 11, 13, 13, 17

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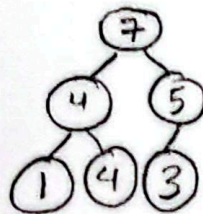
Insert 7



0	1	2	3	4	5	6
	5	4	3	1	4	7

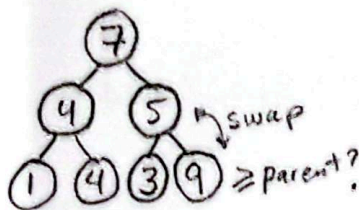


0	1	2	3	4	5	6
	5	4	7	1	4	3

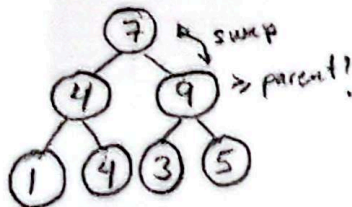


0	1	2	3	4	5	6
	7	4	5	1	4	3

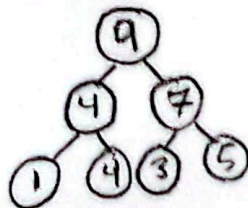
Insert 9



0	1	2	3	4	5	6	7
	7	4	5	1	4	3	9



0	1	2	3	4	5	6	7
	7	4	9	1	4	3	5

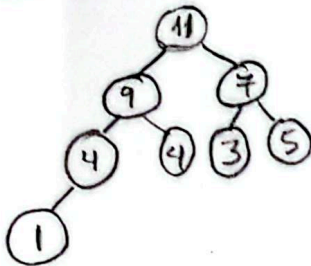
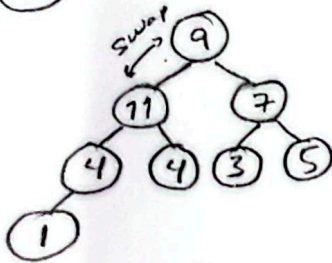
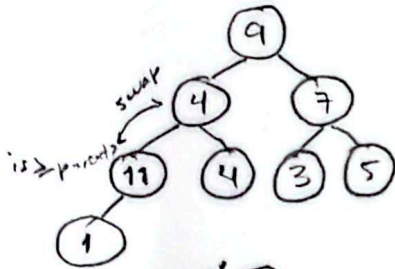
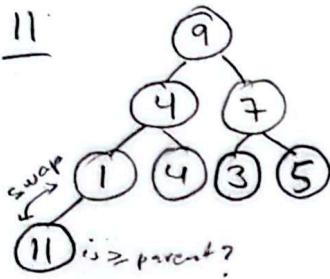


0	1	2	3	4	5	6	7
	9	4	7	1	4	3	5

Q1.a)

2050

Insert 11



0	1	2	3	4	5	6	7	8
	9	4	7	1	4	3	5	11

0	1	2	3	4	5	6	7	8
	9	4	7	11	4	3	5	1

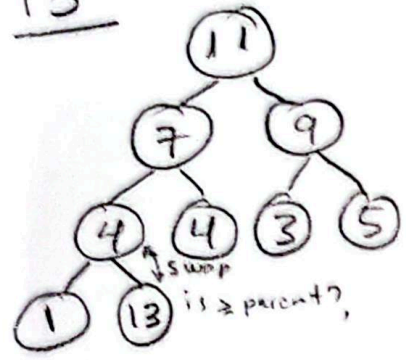
0	1	2	3	4	5	6	7	8
	9	11	7	4	4	3	5	1

0	1	2	3	4	5	6	7	8
	11	9	7	4	4	3	5	1

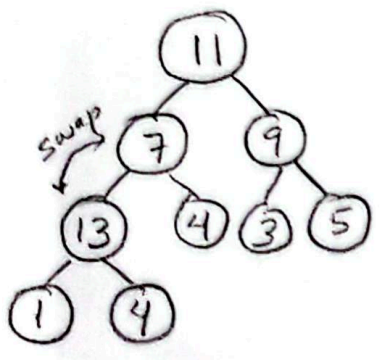
Q1. a)

Jose

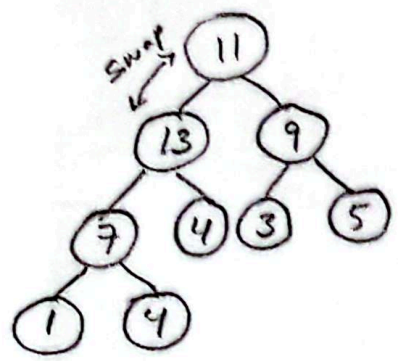
Insert 13



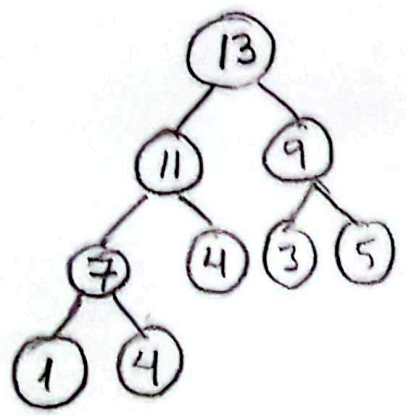
0	1	2	3	4	5	6	7	8	9
	11	7	9	4	4	3	5	1	13



0	1	2	3	4	5	6	7	8	9
	11	7	9	13	4	3	5	1	4



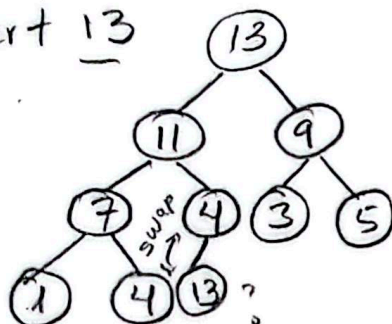
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	11	13	9	7	4	3	5	1	4



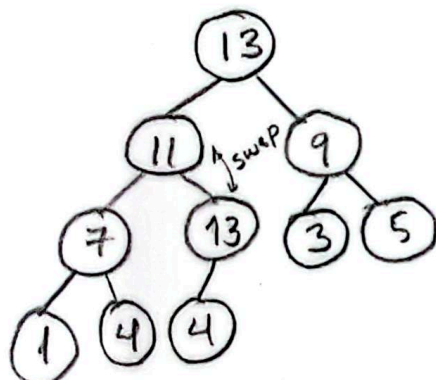
0	1	2	3	4	5	6	7	8	9
	13	11	9	7	4	3	5	1	4

Q1. a)

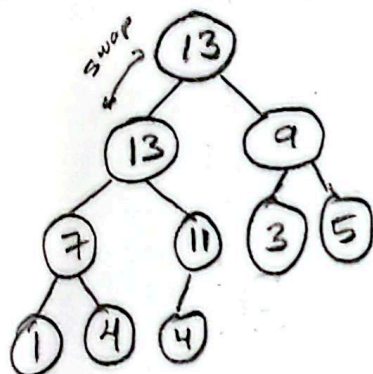
Insert 13



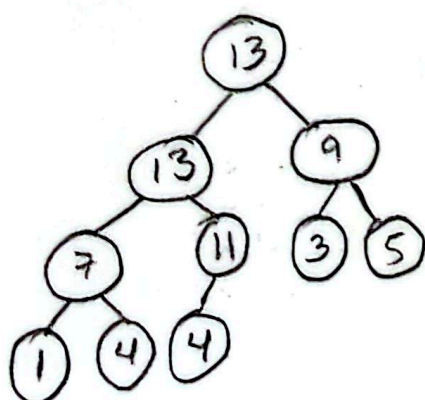
0	1	2	3	4	5	6	7	8	9	10
	13	11	9	7	4	3	5	1	4	13



0	1	2	3	4	5	6	7	8	9	10
	13	11	9	7	13	3	5	1	4	4



0	1	2	3	4	5	6	7	8	9	10
	13	13	9	7	11	3	5	1	4	4

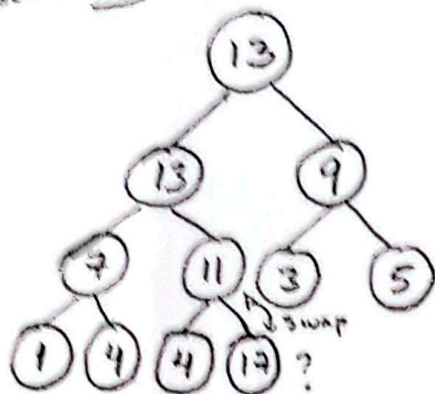


0	1	2	3	4	5	6	7	8	9	10
	13	13	9	7	11	3	5	1	4	4

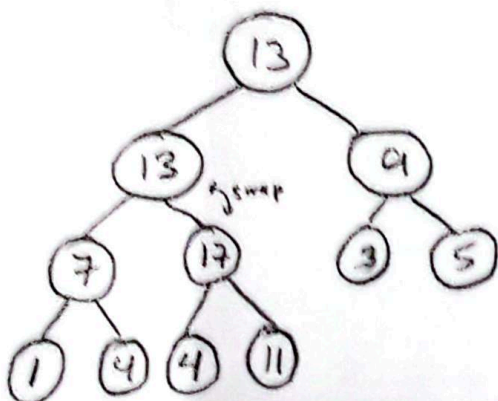
Q2.a)

Jose

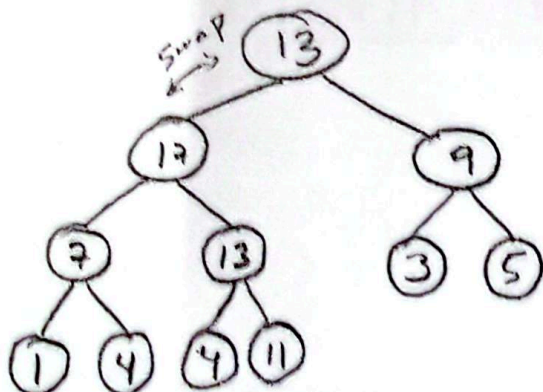
Insert 17



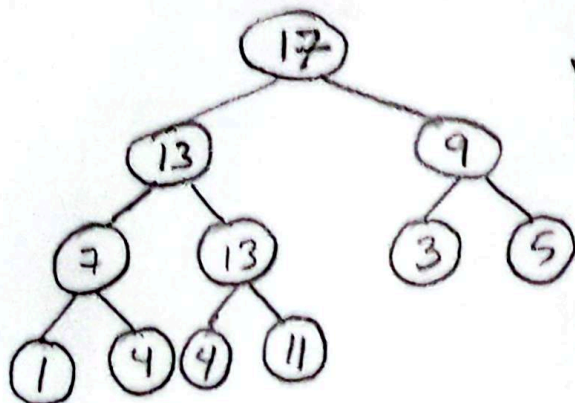
0	1	2	3	4	5	6	7	8	9	10	11
	13	13	9	7	11	3	5	1	4	4	17



0	1	2	3	4	5	6	7	8	9	10	11
	13	13	9	7	17	3	5	1	4	4	11



0	1	2	3	4	5	6	7	8	9	10	11
	13	17	9	7	13	3	5	1	4	4	11



0	1	2	3	4	5	6	7	8	9	10	11
	17	13	9	7	13	3	5	1	4	4	11